Republic of the Philippines
LAGUNA STATE POLYTECHNIC UNIVERSITY
Province of Laguna

ENTERPRISE INFRASTRUCTURE PLAN

NorthBridge Software Solutions — IT Infrastructure Planning for a Startup Company

Week 2 Portfolio Project

ITEP 414 — System Administration and Maintenance

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Role	Junior System Administrator (assigned)
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PART 1 | Company Profile

1.1 Company Overview

Company Name: NorthBridge Software Solutions Inc.

Nature of Business: Custom software development and IT outsourcing for small and mid-sized clients (web applications, business systems, and cloud-hosted solutions).

Year Established: 2026

Office Location (fictional): Unit 4B, TechnoHub Business Center, Calamba Techno Park, Calamba City, Laguna, Philippines

Number of Employees: 20

Office Setup: Single-floor leased office with four functional areas: IT/Development, Human Resources, Finance, and Sales

1.2 Company Vision

To become a trusted regional partner in software innovation, recognized for building reliable, secure, and scalable digital solutions that help small and mid-sized businesses grow with confidence.

1.3 Company Mission

- Deliver high-quality, well-engineered software solutions on time and within budget.
- Build a secure and reliable IT infrastructure that supports continuous business operations.
- Foster a culture of professionalism, continuous learning, and technical excellence among employees.
- Protect client and company data through sound security and administration practices.

1.4 Organizational Structure

NorthBridge Software Solutions follows a flat, function-based organizational structure appropriate for a 20-person startup. The Chief Executive Officer oversees four department heads, each responsible for a functional unit. The IT Department additionally carries the responsibility of maintaining the company's own internal infrastructure, in addition to client-facing development work.

- Chief Executive Officer (CEO) — overall business direction and client relations
  - IT Department Head — software development, systems administration, and internal infrastructure
  - HR Department Head — recruitment, employee relations, and compliance
  - Finance Department Head — budgeting, payroll, and financial reporting
  - Sales Department Head — client acquisition, account management, and business development

1.5 Employee Distribution

Department	Employees	Primary Function
Information Technology	5	Software development, systems/network administration, technical support
Human Resources	4	Recruitment, employee records, training and compliance
Finance	5	Payroll, accounts, budgeting, financial reporting

Department	Employees	Primary Function
Sales	6	Client acquisition, account management, field and remote client visits
TOTAL	20	

PART 2 | Enterprise Hardware Inventory

The hardware inventory below was designed to support 20 employees across four departments, with additional allowances for redundancy (spare devices, backup power, and backup storage) in line with good system administration practice. Quantities reflect one primary device per employee, with department-specific adjustments — for example, Sales staff are issued laptops rather than desktops to support client visits, and the IT Department receives higher-specification machines to support development and virtualization workloads.

Asset ID	Hardware	Qty	Department	Purpose / Justification
AST-001	Desktop Computer	5	Information Technology	Development workstations for coding, testing, and virtualization; require higher CPU/RAM specs
AST-002	Desktop Computer	4	Human Resources	Standard office workstations for records management and employee documentation
AST-003	Desktop Computer	5	Finance	Workstations for accounting software, payroll processing, and financial reporting
AST-004	Laptop	6	Sales	Portable devices for client meetings, presentations, and remote/field work
AST-005	Laptop	2	IT / Management	Spare and administrative laptops for the IT head and system administrator
AST-006	Server (Application/File)	1	IT Department	Hosts internal business applications, file sharing, and development/staging environments
AST-007	Server (Domain/Backup)	1	IT Department	Directory services, authentication, and scheduled backup jobs
AST-008	Router	1	Company-wide	Gateway device connecting the internal network to the ISP
AST-009	Managed Switch	2	Company-wide	Core and distribution switching for wired connectivity across departments
AST-010	Network Printer	3	Shared (IT/Dev, HR/Finance, Sales)	Shared printing, scanning, and document reproduction per office zone
AST-011	UPS (Uninterruptible Power Supply)	3	Server Room / Network Rack	Power backup to prevent data loss and hardware damage during outages
AST-012	Wireless Access Point	2	Company-wide	Wi-Fi coverage across the single-floor office for laptops and mobile devices
AST-013	NAS Storage	1	IT Department	Centralized file storage and scheduled backup repository
AST-014	External Backup Drive	3	IT Department	Offsite/rotating backup copies for disaster recovery
AST-015	Monitor	20	Company-wide	One monitor per desktop/laptop user; dual-monitor setup provided for IT and Finance staff

2.1 Justification Summary

- Desktops vs. Laptops: Desktops are assigned to department staff who work primarily on-site (IT, HR, Finance), providing better price-to-performance for fixed workstations. Laptops are reserved for Sales, whose role requires mobility.
- Dual Servers: A two-server setup (application/file server and domain/backup server) separates production workloads from authentication and backup duties, reducing the risk of a single point of failure.
- Redundant UPS Units: Three UPS units protect the server rack, network rack, and shared printer/workstation cluster, minimizing downtime and hardware damage from power interruptions — a common issue in Philippine business environments.
- NAS + External Drives: Combines fast on-site recovery (NAS) with offsite/rotating backup media (external drives) to satisfy the 3-2-1 backup principle.

PART 3 | Enterprise Software Inventory

The software stack balances licensed productivity tools with free and open-source utilities to control startup costs while meeting the operational needs of development, administration, and end-user departments.

Software	Version	License	Purpose
Windows 11 Pro	23H2	OEM / Volume License	Primary operating system for all desktop and laptop endpoints; supports domain join, BitLocker, and Group Policy management
Ubuntu Server	22.04 LTS	Free / Open Source	Operating system for the application/file server; long-term support and stability for hosting internal services
Microsoft Office	365 Business Standard	Subscription (per user)	Word processing, spreadsheets, presentations, and email/collaboration for all departments
Visual Studio Code	1.9x (latest stable)	Free / Open Source	Primary code editor for the development team
Git	2.4x (latest stable)	Free / Open Source	Version control for all software development projects
GitHub Desktop	3.x (latest stable)	Free	Simplified Git client for repository management and collaboration
VirtualBox	7.x	Free / Open Source	Local virtualization for testing multiple OS environments and isolated development sandboxes
Google Chrome	Latest stable	Free	Standard web browser for all staff; required for cloud-based business applications
Microsoft Defender	Built-in (Windows 11)	Included with Windows 11 Pro	Baseline endpoint antivirus and threat protection on all Windows machines
AnyDesk	8.x	Free (Business tier optional)	Remote desktop access for IT support and remote troubleshooting
7-Zip	23.x	Free / Open Source	File compression and archive management

3.1 Software Rationale

- Windows 11 Pro is standardized on all endpoints instead of the Home edition because Pro supports domain join, Group Policy, and BitLocker encryption — all required for centralized management and data protection.
- Ubuntu Server was selected for the file/application server due to its stability, low licensing cost, and wide community/enterprise support for hosting Linux-based services.
- Development tools (VS Code, Git, GitHub Desktop, VirtualBox) are limited to the IT Department's five workstations, keeping licensing and support overhead low for the remaining departments.
- Microsoft Defender is used as the baseline antivirus since it is already included with Windows 11 Pro at no additional cost; Part 7 discusses whether a supplemental endpoint security product is recommended.

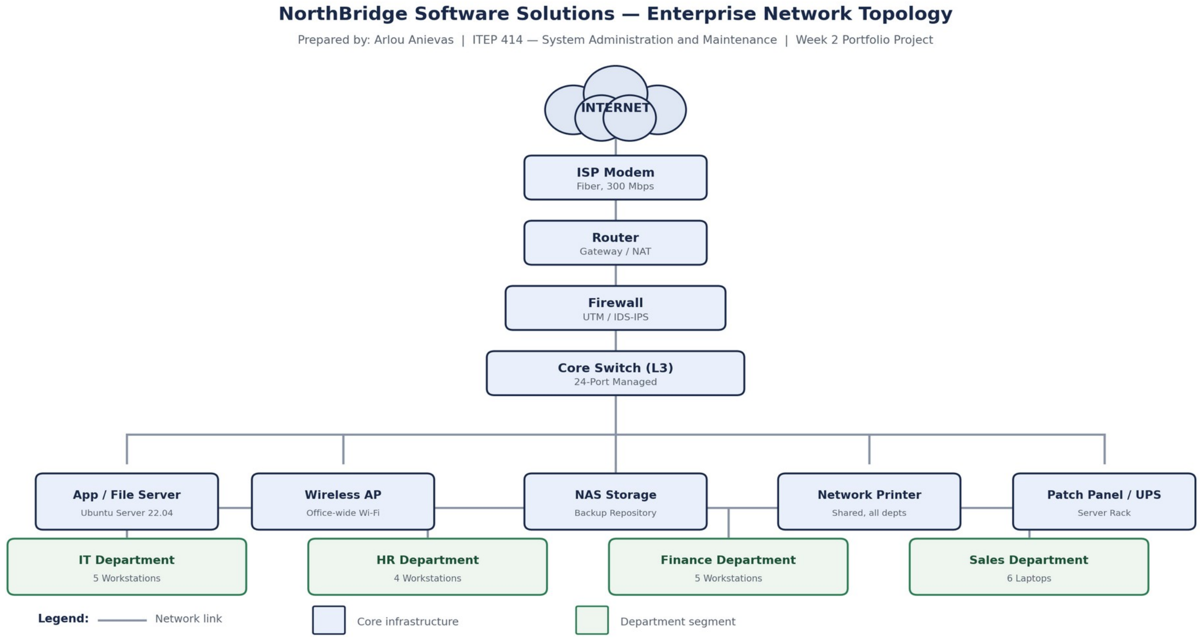
PART 4 | Enterprise Network Inventory

The networking equipment list establishes a segmented, manageable network capable of supporting 20 users, four department VLANs, and future growth without a major redesign.

Item	Specification	Qty	Purpose
ISP Modem	DOCSIS/Fiber ONT, Gigabit-capable	1	Terminates the ISP fiber connection and hands off to the router
Router	Dual-WAN capable, Gigabit ports	1	Provides NAT/routing between the internal network and the internet
Firewall	UTM appliance with IDS/IPS	1	Perimeter security, traffic filtering, and intrusion detection/prevention
Core Switch	24-port Layer 3 managed switch	1	Central switching backbone with VLAN segmentation for departments
Access Switch	24-port Layer 2 managed switch	1	Secondary switch for additional wired drops and expansion
Wireless Access Point	Wi-Fi 6, dual-band, PoE	2	Provides wireless coverage across the single-floor office
Patch Panel	24-port CAT6 patch panel	1	Organizes and terminates structured cabling in the server rack
CAT6 Cable	305 m box, solid copper	2 boxes	Structured cabling for all wired workstation and device drops
RJ45 Connectors	CAT6-rated, box of 100	2 boxes	Termination of CAT6 cable runs at both ends

PART 5 | Enterprise Network Diagram

The diagram below illustrates the logical network topology for NorthBridge Software Solutions, from the internet connection down to individual department segments. The full-resolution diagram (PNG and PDF exports) is included in the diagrams/ folder of the accompanying GitHub repository, as prepared in Draw.io.



5.1 Topology Description

- Internet connectivity enters through the ISP modem, which hands off to the router for NAT and default gateway services.
- A dedicated firewall appliance sits between the router and the core switch, providing perimeter filtering and intrusion detection/prevention before any traffic reaches internal devices.
- The Layer-3 core switch distributes traffic to the server room (application/file server, NAS storage, patch panel/UPS), the shared network printer, and the wireless access point.
- Each department — IT, HR, Finance, and Sales — connects as a logically separated segment (VLAN) off the core switch, isolating department traffic while still allowing controlled access to shared resources such as the server and printer.
- This layered design (Internet → Modem → Router → Firewall → Core Switch → Access Layer) follows the standard defense-in-depth approach used in small-business network design.

PART 6 | System Administration Roles

This section researches four IT roles that a growing company such as NorthBridge Software Solutions will eventually need, either as in-house staff or outsourced specialists, as the organization scales beyond its initial 20-person team.

6.1 Helpdesk Technician

Responsibilities:

- Serves as the first point of contact for employee IT issues (hardware faults, software errors, account lockouts, connectivity problems).
- Logs, tracks, and escalates tickets that cannot be resolved at the first level of support.
- Performs basic hardware setup, software installation, and account provisioning for new employees.

Skills:

- Strong troubleshooting and customer-service communication skills.
- Working knowledge of Windows and basic networking concepts (IP addressing, connectivity testing).
- Ability to document recurring issues and follow standard operating procedures.

Common Tools:

- Ticketing systems (e.g., Freshservice, Zendesk, Jira Service Management)
- Remote support tools (e.g., AnyDesk, TeamViewer)
- Basic diagnostic utilities (ping, ipconfig/ifconfig, Event Viewer)

Certifications:

- CompTIA A+
- CompTIA IT Fundamentals (ITF+)
- Microsoft 365 Certified: Fundamentals

6.2 Network Administrator

Responsibilities:

- Designs, configures, and maintains the organization's routers, switches, firewalls, and wireless infrastructure.
- Monitors network performance and availability, and troubleshoots connectivity or latency issues.
- Implements network security measures such as VLAN segmentation, firewall rules, and VPN access.

Skills:

- Solid understanding of TCP/IP, routing, switching, and VLAN design.
- Familiarity with firewall and VPN configuration.
- Network monitoring and capacity-planning skills.

Common Tools:

- Cisco IOS / packet analysis tools (e.g., Wireshark)
- Network monitoring platforms (e.g., PRTG, Zabbix, SolarWinds)
- Firewall management consoles (e.g., pfSense, Fortinet)

Certifications:

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+
- Fortinet NSE (or equivalent firewall-vendor certification)

6.3 Linux System Administrator

Responsibilities:

- Installs, configures, and maintains Linux-based servers (such as the company's Ubuntu Server file/application server).
- Manages user accounts, file permissions, services, and scheduled jobs (cron).
- Applies security patches, monitors system logs, and maintains server uptime.

Skills:

- Command-line proficiency (Bash scripting, package management, systemd).
- Understanding of file permissions, services, and process management.
- Backup, monitoring, and basic security-hardening practices.

Common Tools:

- Bash / shell scripting
- SSH for remote administration
- Configuration and monitoring tools (e.g., cron, systemd, Nagios)

Certifications:

- Red Hat Certified System Administrator (RHCSA)
- Linux Professional Institute Certification (LPIC-1)
- CompTIA Linux+

6.4 Cloud Administrator

Responsibilities:

- Manages the organization's cloud-hosted resources (virtual machines, storage, and backup services) as the company migrates workloads off-site.
- Monitors cloud costs, availability, and performance, and applies scaling adjustments as needed.
- Implements identity and access management (IAM) policies and cloud security best practices.

Skills:

- Familiarity with at least one major cloud platform (AWS, Microsoft Azure, or Google Cloud).
- Understanding of cloud networking, storage tiers, and cost optimization.
- Scripting/automation skills (e.g., using cloud CLI tools or infrastructure-as-code).

Common Tools:

- Cloud provider consoles and CLI tools (AWS CLI, Azure CLI)
- Infrastructure-as-code tools (e.g., Terraform)
- Cloud monitoring dashboards (e.g., CloudWatch, Azure Monitor)

Certifications:

- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Azure Administrator Associate
- Google Associate Cloud Engineer

6.5 How These Roles Work Together

Within a single organization such as NorthBridge Software Solutions, these four roles form a layered support structure rather than four isolated jobs. The Helpdesk Technician handles day-to-day, front-line employee issues and filters out routine problems so that specialists are not pulled away from higher-value work; issues beyond first-level scope are escalated to the Network Administrator or Linux System Administrator depending on whether the root cause is connectivity-related or server-related. The Network Administrator keeps the underlying infrastructure — routers, switches, firewall, and Wi-Fi — reliable and secure, which is the foundation every other role depends on. The Linux System Administrator keeps the company's servers, internal applications, and backups running, working closely with the Network Administrator to make sure server traffic is properly segmented and protected. As the company grows and begins hosting workloads off-site, the Cloud Administrator extends this same discipline into hosted environments, coordinating with the Linux System Administrator on hybrid backup strategies and with the Network Administrator on secure site-to-cloud connectivity. Together, these four roles ensure that infrastructure is supported at every layer: the user, the network, the on-premises server, and the cloud.

PART 7 | Infrastructure Recommendations

7.1 Internet Service Provider

A business-grade fiber connection (minimum 300 Mbps symmetrical) from an established Philippine enterprise ISP (e.g., PLDT Enterprise, Globe Business, or Converge ICT Business) is recommended over a residential plan. Business-grade fiber includes a Service Level Agreement (SLA) with guaranteed uptime and priority restoration, which is essential since the company's development, file server, and client-facing work depend on continuous connectivity. A secondary backup connection (e.g., a mobile broadband failover) is recommended once the budget allows, to avoid a single point of failure.

7.2 Server Specifications

For the initial 20-employee scale, a single rack-mount server (or well-specified tower server) running Ubuntu Server 22.04 LTS is sufficient, with a second unit reserved for backup/domain services as listed in Part 2. Recommended baseline specifications:

- Processor: Multi-core server-grade CPU (e.g., Intel Xeon E-2300 series or equivalent) to support virtualization and concurrent services.
- Memory: Minimum 32 GB ECC RAM, expandable, to comfortably run file sharing, internal applications, and light virtualization.
- Storage: RAID 1 or RAID 10 configuration using SSDs for performance and redundancy, sized around 2 TB usable capacity with room for growth.
- Power & Cooling: Dual power supplies where budget allows, connected to a dedicated UPS, in a ventilated server rack or closet.

7.3 Backup Strategy

A 3-2-1 backup strategy is recommended: at least three copies of critical data, stored on two different media types, with one copy kept offsite. In practice, this means daily automated backups from the file/application server to the on-site NAS, weekly rotation of an external backup drive stored offsite (or with a designated employee), and, as the budget allows, a cloud-based backup service for the most critical business and client data. Backup jobs should be tested periodically through sample file restores to confirm they are actually recoverable, not just scheduled.

7.4 Security Recommendations

- Segment the network into VLANs per department, restricting cross-department access to only what is operationally necessary.
- Deploy the UTM firewall with IDS/IPS enabled, and keep firmware and signature databases updated.
- Enable centralized logging and periodic log review to detect unusual access patterns early.
- Apply the principle of least privilege for file, application, and server access — employees should only have access to what their role requires.
- Enforce automatic OS and application patching schedules across all endpoints and the server.

7.5 Antivirus

Microsoft Defender (already included with Windows 11 Pro) is an adequate baseline for individual endpoints at no additional licensing cost. However, as the company grows, a centrally managed endpoint protection platform (e.g., Microsoft Defender for Business, or a comparable managed EDR product) is recommended so that the IT Department can monitor threat status across all devices from a single console rather than checking each machine individually.

7.6 Password Policy

- Minimum password length of 12 characters, combining upper/lowercase letters, numbers, and symbols.
- Mandatory multi-factor authentication (MFA) for email, cloud services, and remote access (e.g., AnyDesk, VPN).
- Password changes required only when there is evidence of compromise, in line with current NIST guidance, rather than arbitrary periodic resets which tend to encourage weak password reuse.
- Use of a company-approved password manager to avoid password reuse across systems.
- Immediate deactivation of credentials upon employee offboarding.

7.7 Expansion Plan

The current design anticipates growth beyond the initial 20 employees. The core switch and network cabling are sized with spare ports and drops for near-term hires. As headcount grows past roughly 35–40 employees, or as additional office space is leased, the recommended next steps are: (1) add a second access switch and additional wireless access points for coverage, (2) migrate select workloads (e.g., backups, email, and non-sensitive applications) to a cloud provider to reduce on-premises server load, and (3) formally bring on a dedicated Network Administrator and Cloud Administrator, rather than having these responsibilities absorbed informally by the existing IT staff.

PART 8 | Personal Reflection

Prepared by Arlou Anievas

Working through this Enterprise Infrastructure Plan gave me a much clearer picture of what a System Administrator actually does before a single cable is plugged in or a single server is powered on. Coming into this project, I assumed that "system administration" mostly meant fixing problems after they happened — troubleshooting a broken printer, resetting a password, or restarting a server. Building this plan from scratch, for a company with absolutely nothing in place, showed me that the real work happens earlier than that: in understanding the business first, and only then translating those needs into hardware, software, and network decisions.

The most challenging part of this project was Part 2 and Part 4, deciding on realistic hardware and networking quantities and being able to justify each one. It was tempting to just list generic numbers, but I had to keep asking myself questions like: why does Sales need laptops instead of desktops, why does the server room need three UPS units instead of one, and why should HR and Finance be separated from IT on the network. Working through those questions forced me to think less like someone filling out a checklist and more like someone who would actually be accountable if the network went down or a department lost access to shared files. Designing the network diagram was also demanding in a different way — translating a list of equipment into a logical topology that actually made sense, with traffic flowing correctly from the internet down to each department, took several attempts before the layout felt clean and defensible.

This project reinforced why planning has to come before deployment. If NorthBridge Software Solutions had simply bought computers and a router without first mapping out departments, security needs, and growth plans, the company would likely end up with a network that is hard to secure, hard to expand, and expensive to fix later. Planning first — profiling the company, inventorying needs, designing the topology, and only then writing recommendations — means that every purchase and configuration decision has a reason behind it, and that the infrastructure can grow with the business instead of constantly being rebuilt.

Overall, this project helped me become a better System Administrator in training by connecting technical knowledge to business context. I already had some exposure to networking concepts and Linux basics from previous coursework, but this was the first time I had to combine hardware selection, software licensing, network design, security policy, and cost-conscious recommendations into a single coherent plan aimed at a non-technical audience — an IT Manager or company executive. That combination of technical accuracy and clear communication is, I now realize, at the core of what makes system administration valuable to an organization, and it is a skill I intend to keep practicing throughout the rest of this course and into my future internship and career.

References

The following sources informed the certification, security, and role-based research used throughout this plan, particularly Parts 6 and 7. Full citation details and any additional reading are also listed in the references/ folder of the accompanying GitHub repository.

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